



MARCH 9 – CODING WRAP UP



WHERE ARE WE?

RAG (RETRIEVED AUGMENTED GENERATION)

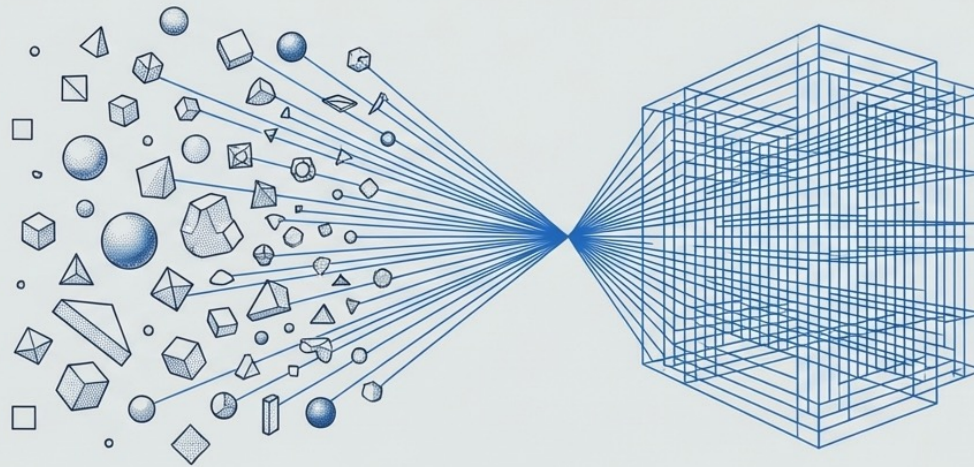
It's a strategy to improve relevance and accuracy of results of querying LLMs.

We are talking about in the context of literature search tools.

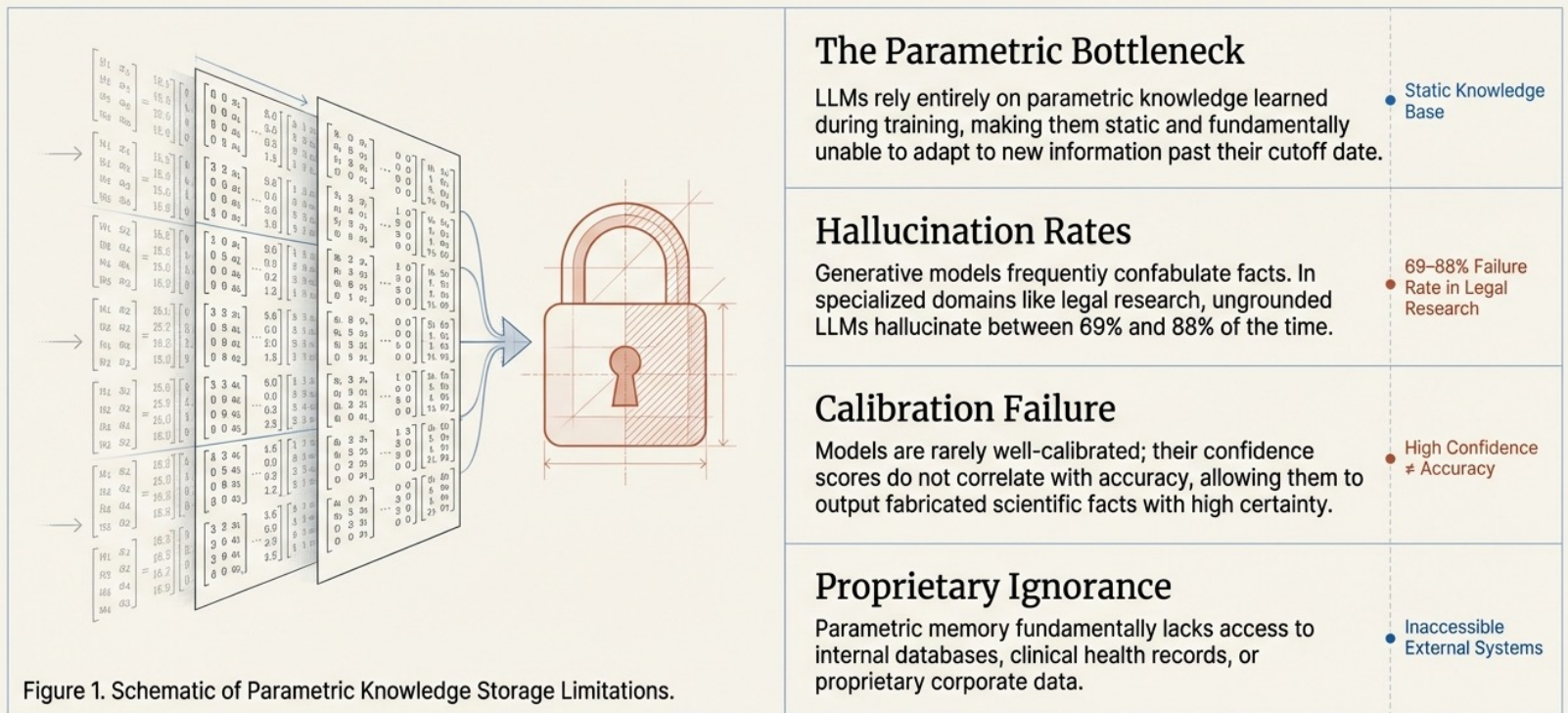
RAG IN 1 SLIDE

- Need for specialized knowledge to improve results from generic LLMs
- Can use enhanced prompting (we know it's efficient because we can @files!); in many cases, not practical if corpus of knowledge is too large
- Augmenting = access to a chunked, embedded vector space of relevant content
- At query time, the most relevant chunks are retrieved (some measure of similarity is needed)
- They are added to the context window to improve results

Beyond the Weights: The Evolution of Retrieval-Augmented Generation



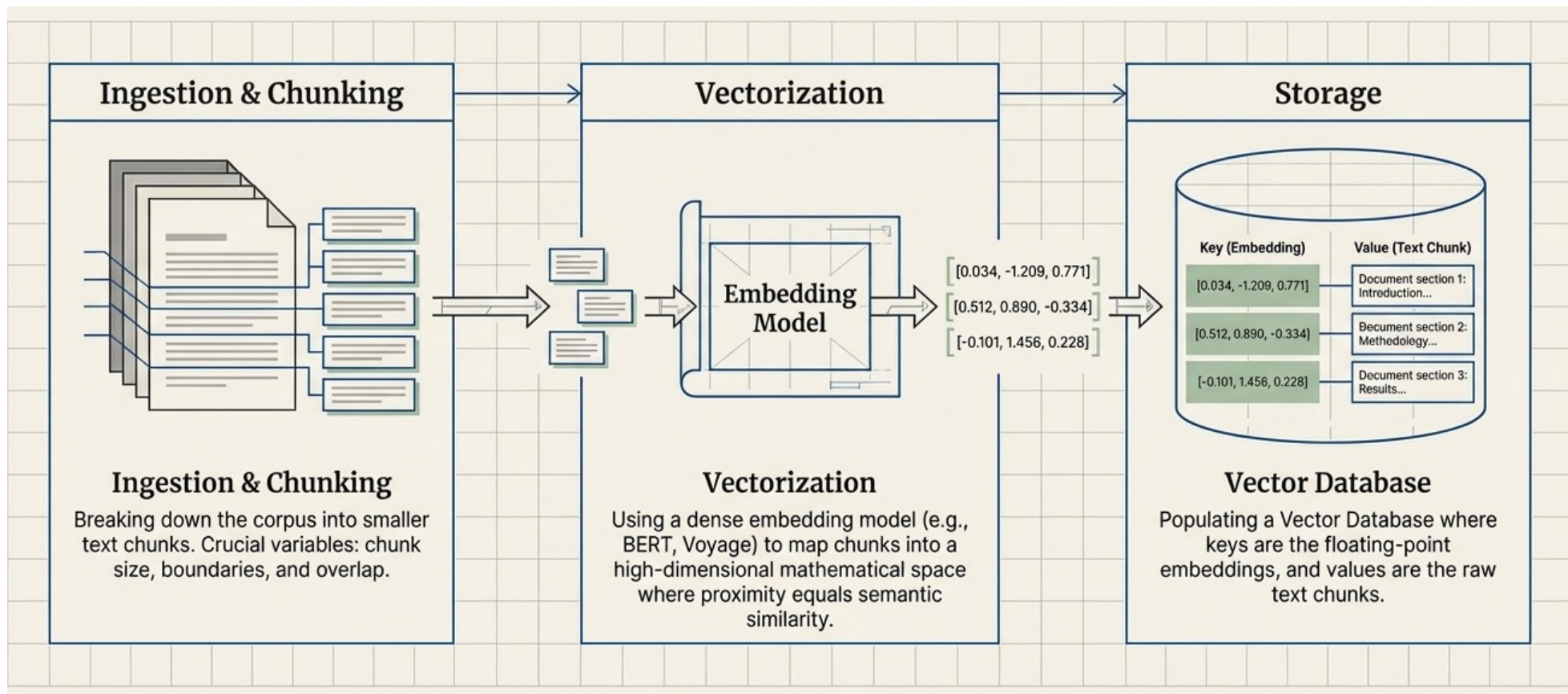
The limits of storing facts in feedforward weights



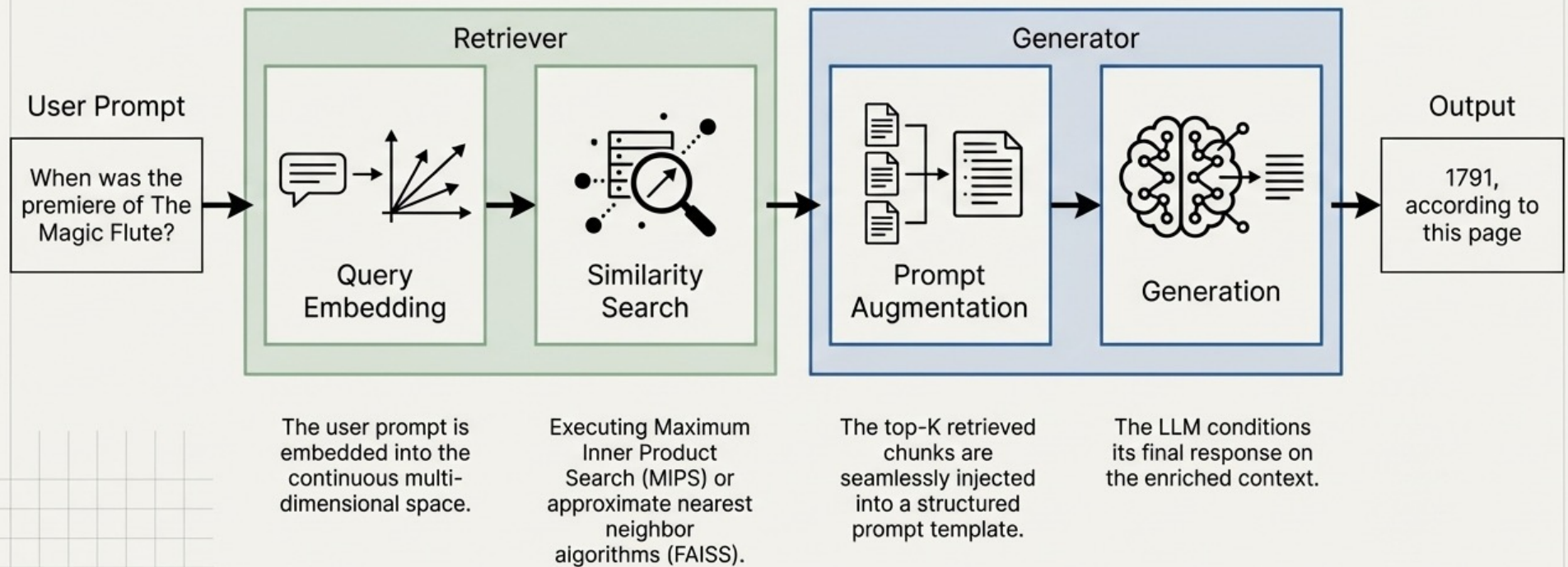
THE COMPONENTS OF RETRIEVAL- AUGMENTED GENERATION (RAG)

- The RAG Paradigm combines a pre-trained sequence-to-sequence (seq2seq) model with a dense vector index of external knowledge.
- Non-Parametric Memory (Retriever): A dynamic knowledge base that allows access to updated information.
- Parametric Memory (Generator): A pre-trained model that uses its weights and the retrieved documents to generate responses.

Transforming a corpus from raw text to vector space



THE RUNTIME INFERENCE PIPELINE



IMPROVEMENTS ON VANILLA RAG

Query expansion (improve quality of query and/or match to information corpus)

Graph RAG: Break corpus into a graph: entity 1, entity 2, and relationship between them
(say for example you want to ask about the relationship between two characters, Romeo and Juliet;
this improves information encoding)

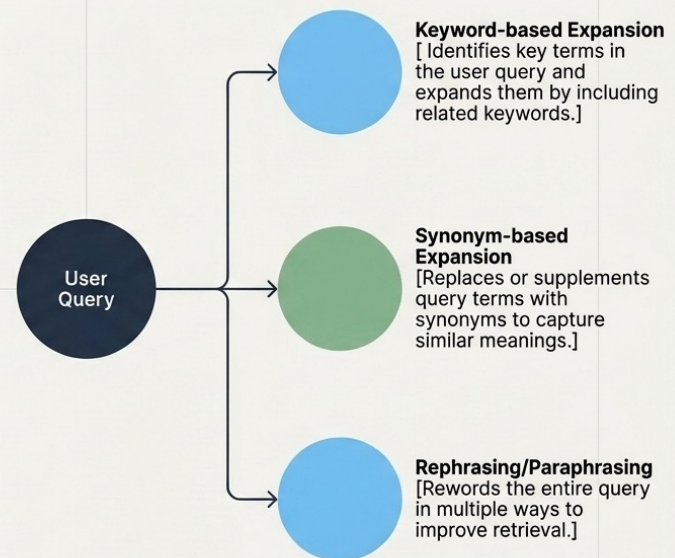
Hybrid search: Embedding + BEM25 or TFIDF (combine semantic similarity with token/keyword matching algorithms
to improve quality of retrieved documents)

Re-ranking: retrieve a larger subset of relevant docs, then re-rank those as context through an LLM query

Fine tuning: fine tune a small language model to be specialized on a topic
(add more relevant training set and re-train; see for example [SPECTER](#), Figure 2,
trained only on academic text). The result is better embeddings for RAG

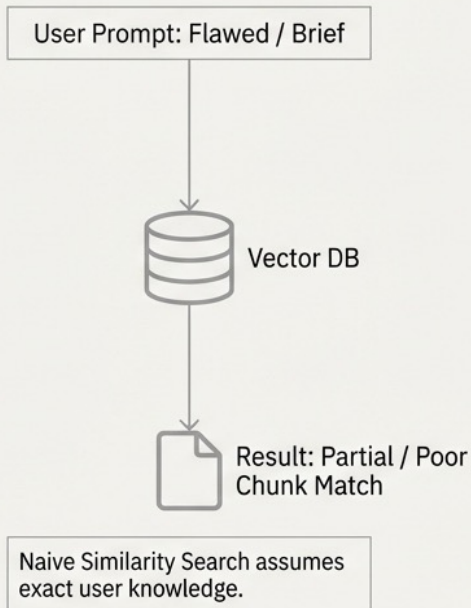
Advanced RAG Pipeline Optimization

Query Expansion & Agentic
Reformulation

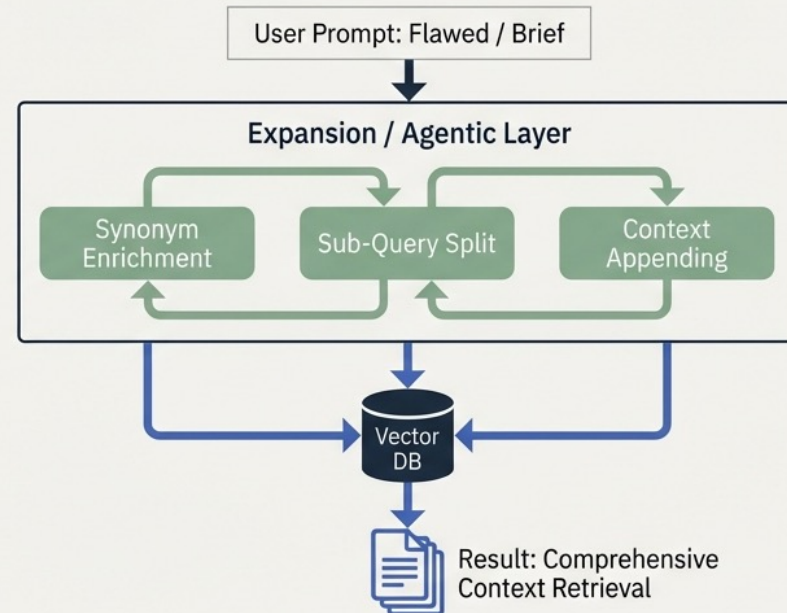


The Intervention Layer: Bridging Human Shorthand and Semantic Space

Standard RAG (Naive Process)



Expanded RAG (Advanced Process)



The Translation Gap

Query expansion bridges the gap between brief human inputs and the strict mathematical realities of semantic vector space.

Agentic Reformulation

Autonomous AI agents intercept the query, enrich it with synonyms, formulate targeted search variants, and iteratively re-retrieve.

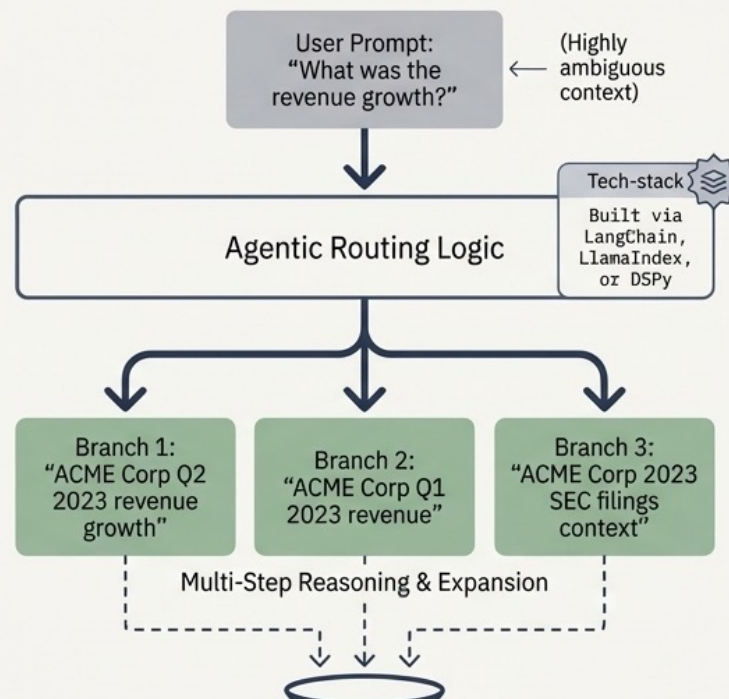
Multi-hop Architecture

Initial baseline documents are retrieved and appended back to the original query to form a for a highly accurate second retrieval stage.

The Anatomy of an Expanded Query

Stacking Benefits

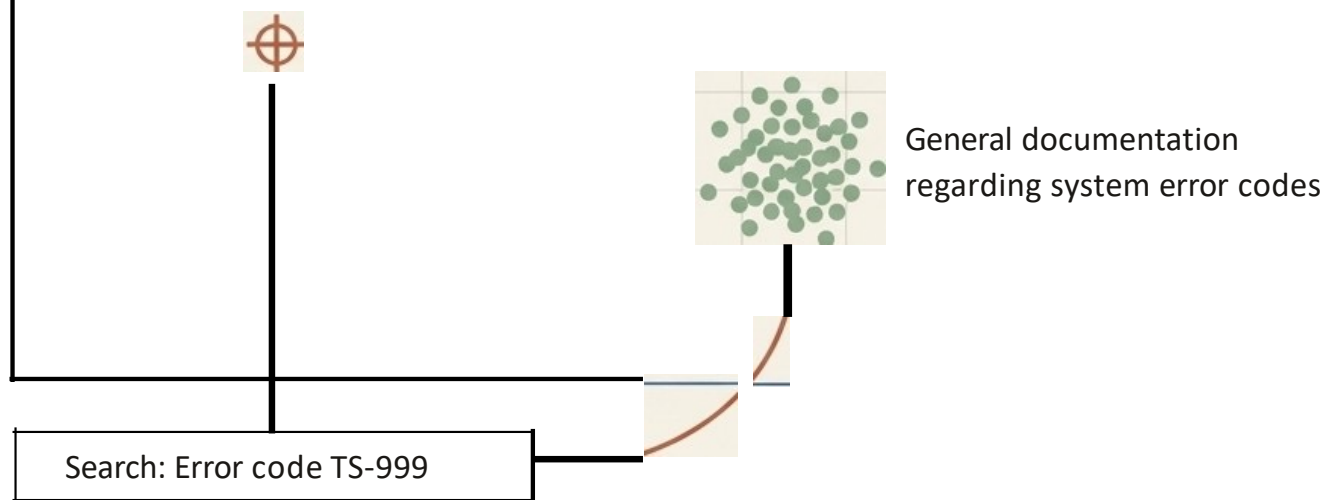
Query Expansion acts as a multiplier. By casting a wider, context-rich net, it provides a superior foundation of chunks for downstream Hybrid Search (Embeddings + BM25) and Reranking models to filter.



Contextual Iteration

Allows the LLM to handle complex tasks—comparing cross-document information and asking follow-up questions—purely dynamically, without costly retraining of underlying model weights.

THE SEMANTIC BLINDSPOT OF DENSE EMBEDDINGS



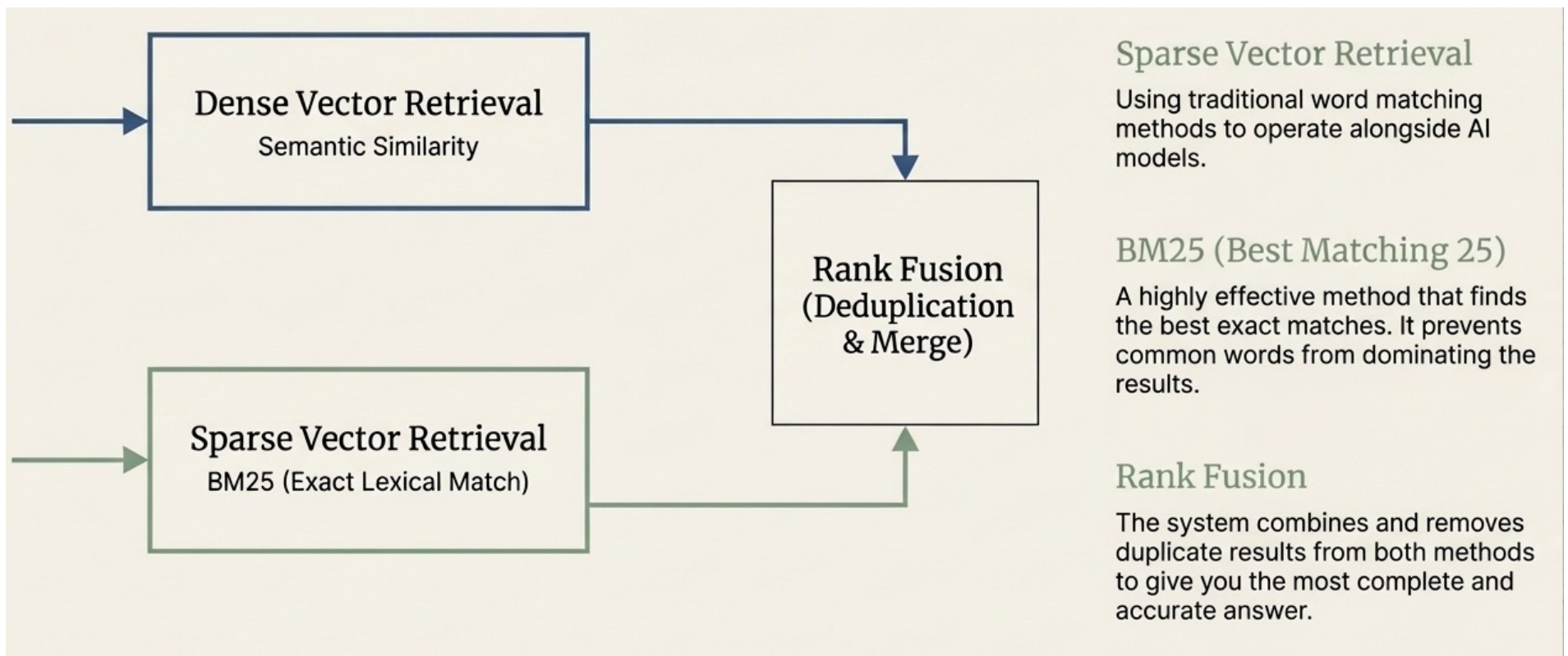
Embedding models are optimized to capture broad semantic relationships and conceptual proximity.

Because they encode meaning, they routinely fail to capture exact syntax, unique scientific identifiers, proprietary IDs, or specific error codes.

The Empirical Example

A query searches for 'Error code TS-999'. The model retrieves general error documentation, but entirely misses the specific string 'TS-999'.

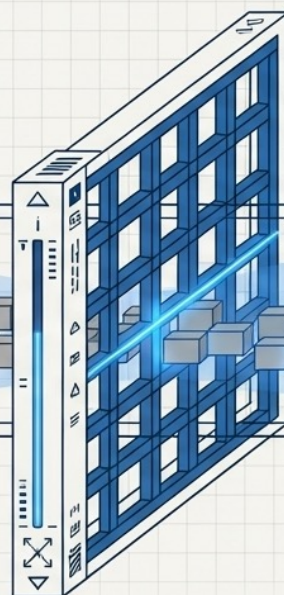
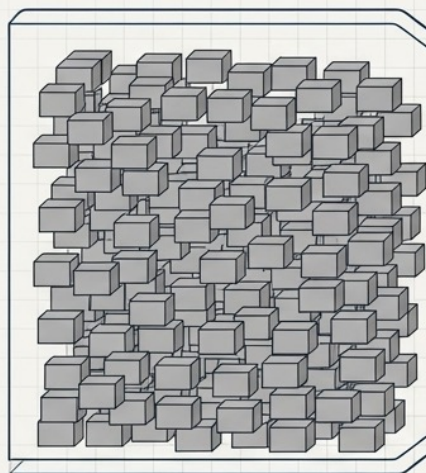
HYBRID SEARCH BALANCES MEANING AND EXACT SYNTAX



High-precision filtering via Reranking

HIGH-PRECISION FILTERING VIA RERANKING

Step 1 (Wide Net): Initial retrieval fetches top-150 chunks using **Hybrid Search**.



Step 3 (Narrow Filter): The system filters down to the highest-scoring **top-20 chunks** to pass to the generator, drastically reducing context distraction.



Step 2 (Cross-Encoder): A specialized reranking model scores each retrieved chunk sequentially based on strict relevance to the specific query.

Note: Standard k-NN retrieval struggles with precision at scale. Reranking introduces a secondary, high-fidelity filter.

Transcending chunk boundaries with Graph RAG

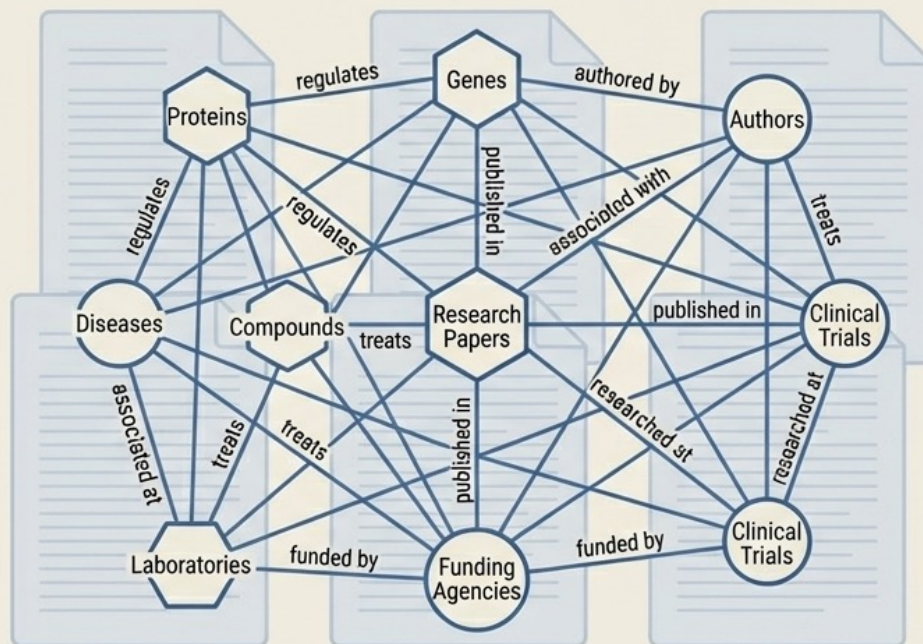
Standard RAG is optimized for explicit question-answering via localized chunks, but fails at corpus-wide summarization.

Knowledge Graphs

Graph RAG utilizes a generative model during ingestion to extract entities and map their relationships across the entire dataset.

Global Synthesis

Allows the system to answer sweeping queries (e.g., "What are the common contradictions in these 500 papers?") by traversing the graph nodes rather than relying on isolated semantic similarity.



Advanced synthesis of scientific knowledge

The architectural evolution from flawed generative weights to advanced external knowledge synthesis is complete.



Empirical Milestone

The PaperQA2 agent, operating without tool or time restrictions, achieves unprecedented **factuality** in **complex** literature searches, exceeding human domain experts on scientific benchmarks (LitQA2).



Contradiction Detection



Successfully identifies an average of **2.34** contradictions per paper in random subsets of biology literature.



70% of those contradictions were subsequently validated by human experts.

PaperQA2: Examples of Synthesis and Contradiction Detection

Synthesis Example



Query: "What are the key mechanisms of drug resistance in EGFR-mutated lung cancer?"

Synthesized Answer:

PaperQA2 identifies primary mechanisms including T790M secondary mutation, MET amplification, and activation of alternative pathways like HER2 or PI3K/AKT, synthesizing findings across 45 papers.

Contradiction Detection Example

Finding: "Paper X (2022) suggests protein A inhibits pathway B."



Contradiction:

However, PaperQA2 highlights that Paper Y (2023) and Paper Z (2021) demonstrate protein A ***activates*** pathway B in different cellular contexts, indicating a context-dependent role.

Original RAG paper

<https://arxiv.org/pdf/2005.11401>

Slides

https://llm.labri.fr/LLM_Course_RAG.pdf

Textbook chapter

<https://web.stanford.edu/~jurafsky/slp3/11.pdf>

Blogs

<https://weaviate.io/blog/introduction-to-rag>

On improvements

<https://www.anthropic.com/engineering/contextual-retrieval>

<https://medium.com/@sahin.samia/query-expansion-in-enhancing-retrieval-augmented-generation-rag-d41153317383>

To read and discuss together (exercise)

<https://arxiv.org/abs/2409.13740> (Paper QA2)

<https://github.com/Future-House/paper-qa>

For next time (Wednesday 3/11):

If possible, install this package: <https://github.com/Future-House/paper-qa>

Think about a mini collection of papers you are familiar with (maybe 2-4) and a couple of research questions (whose answers you know) that you could ask to evaluate the software